



Section 1

AC-CIRCUIT + MACHINERY PRINCIPLE

Production of Induced Force on a Wire

A major effect of a magnetic field on its surroundings is that it induces a force on a current-carrying wire within the field. The induction of a force in a wire by a current in the presence of a magnetic field is the basis of motor action.

The force induced on the conductor is given by:

$$F = i(l \times B)$$

$$F = ilB \sin \theta$$

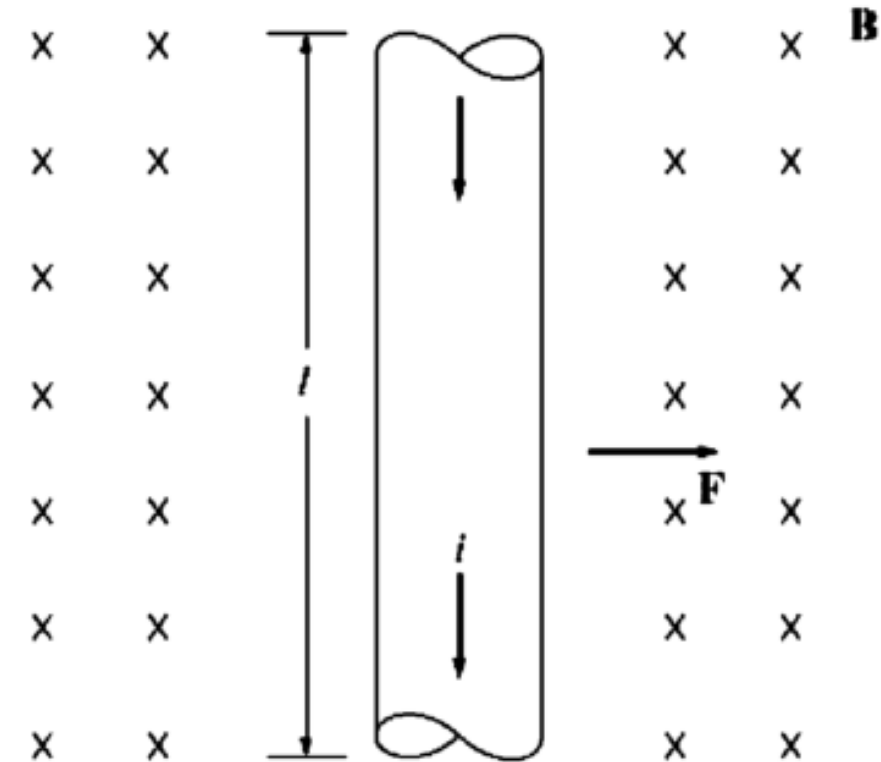
Where:

i : Magnitude of current in wire

l : Length of wire

θ : Is the angle between the wire and the flux density vector

- The direction of the force is given by the right-hand rule: If the index finger of the right-hand points in the direction of the vector $\mathbf{l}$ and the middle finger points in the direction of the flux density vector $\mathbf{B}$, then the thumb points in the direction of the resultant **force** on the wire.



Induced Voltage on a Conductor Moving in a Magnetic Field

There is another major way in which a magnetic field interacts with its surroundings. If a wire with the proper orientation moves through a magnetic field, a voltage is induced in it. The induction of voltages in a wire moving in a magnetic field is fundamental to the operation of all types of generators.

The voltage induced in the wire is given by:

$$e_{ind} = (v \times B) \cdot l$$

Where:

v : Velocity of the wire

B : Magnetic flux density vector

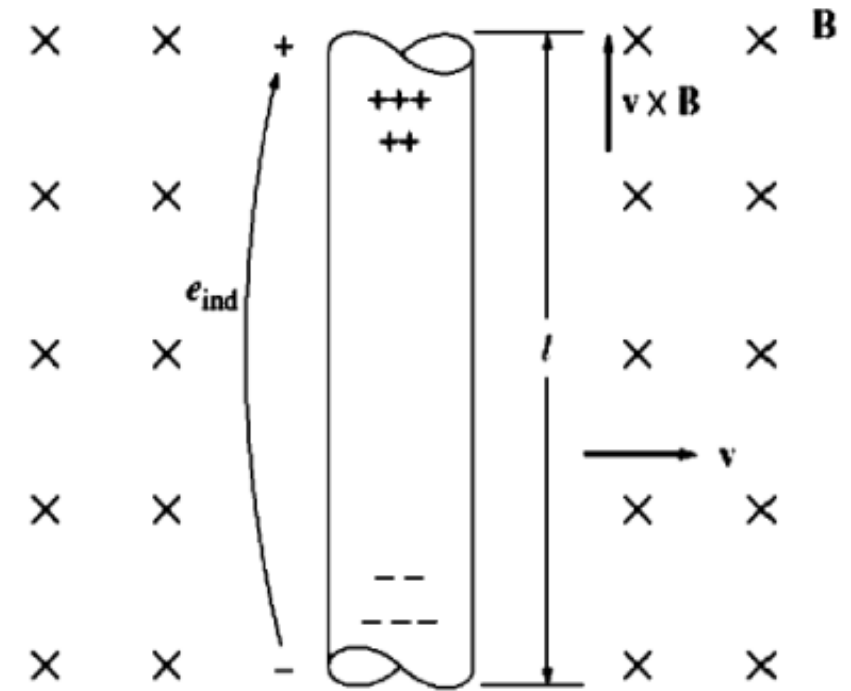
l : Length of conductor in the magnetic field

- Right-Hand Rule:

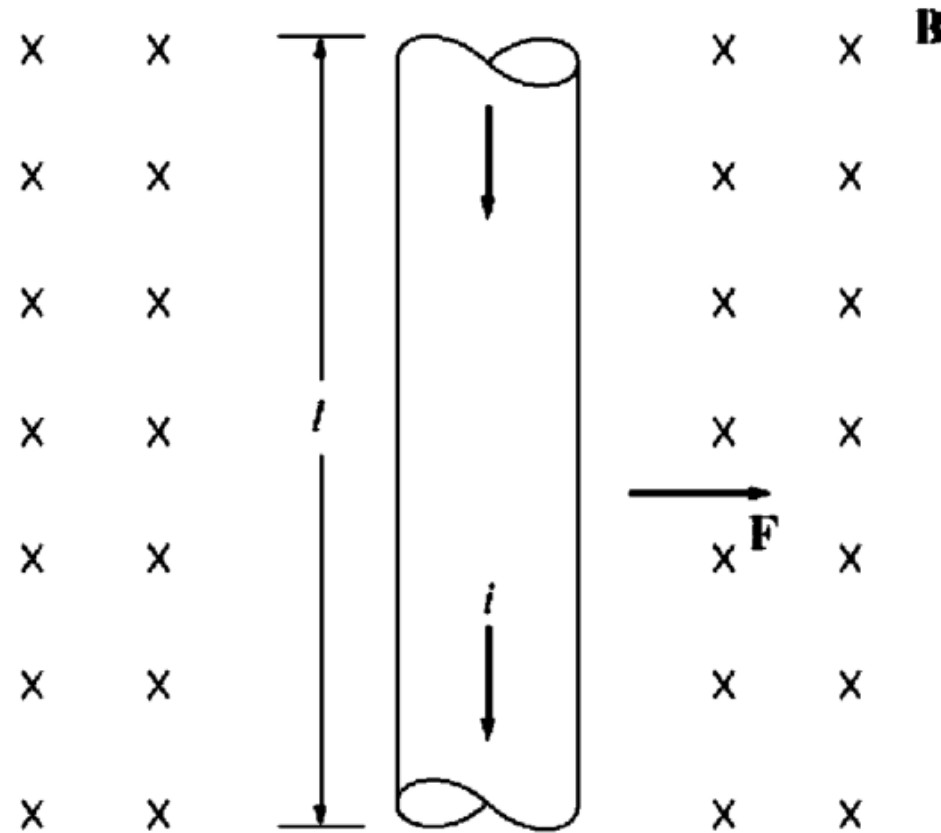
Point your thumb → direction of velocity (v) of the conductor

Point your index finger → direction of magnetic field (B)

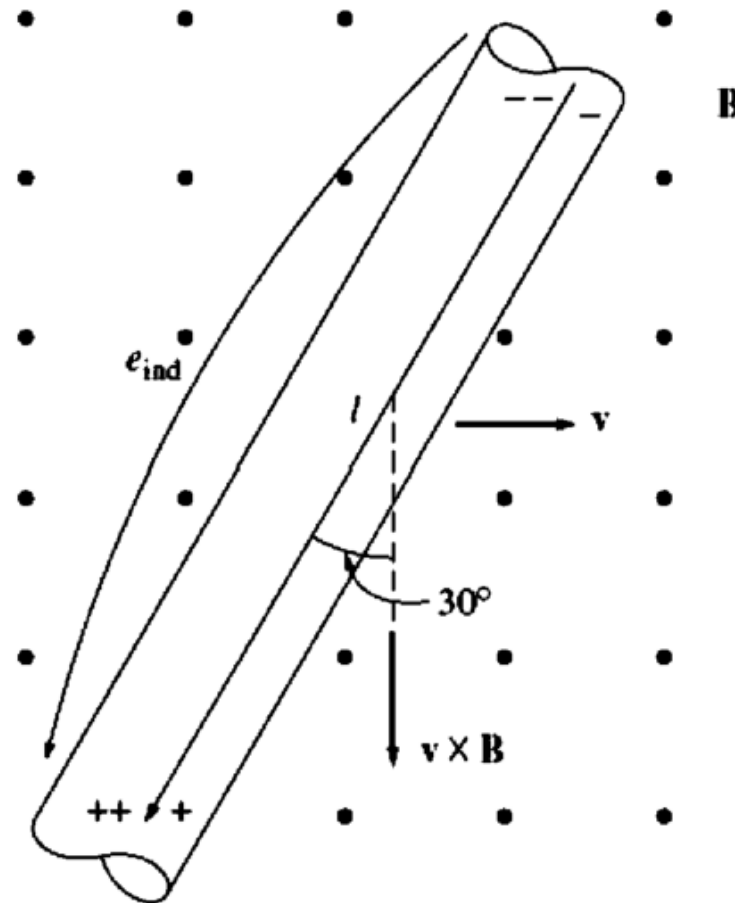
Your middle finger (perpendicular to both)



Example 1: Figure below shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25 T. directed into the page. If the wire is 1.0 m long and carries 0.5A of current in the direction from the top of the page to the bottom of the page. What are the magnitude of the force induced on the wire?



Example 2: Figure below shows a conductor moving with a velocity of 10m/s to the right in a magnetic field. The flux density is 0.5T, out of the page, and the wire is 1.0m in length, oriented as shown. What are the magnitude of induced voltage?



Example 3: A moving wire is in presence of a magnetic field. With the information given in the figure. Determine the magnitude induced voltage in the wire.

